It is not a priori quite clear to the author why exactly the standard framework of measure-theoretic probability (and financial calculus, in particular) is defined as such. This article aims to derive it from common sense and not pull any insights from higher intelligence.

1. Available information

A fundamental concept underlying financial mathematics is the mathematical model of *available information* — the way we encode what we know and what we don't know about a real-world event or process.

1.1. Setup

Imagine a probability experiment that can be "run" and on which "observations" can be made.

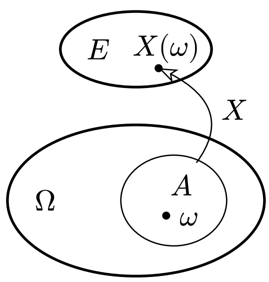
1. A single run fixes (determines) *everything* that can be observed about the experiment.
2. A *run* is commonly denoted by ω, and the set of *all* runs is commonly denoted by Ω.
3. Probabilities are assigned to sensible subsets of Ω, forming a σ-algebra $\mathcal{F} \subset \mathcal{P}(\Omega)$ and a probability measure $\mathbb{P} : \mathcal{F} \to [0, 1]$.
4. In practice, we do not know exactly which run $\omega \in \Omega$ occurred. Instead,
   a. we can only know if the run is *among whole sets* A of runs;
   b. we can usually only "look" at the experiment, and *observe* values of projections $X : \Omega \to E$ onto other sets E.

1.1.1. Events

Such a set of runs A, that we can know if ω lies in, is called *an event* and is only allowed to be in $\mathcal{F}$. Thus it is in the domain of $\mathbb{P}$ and has a well-defined *probability* $\mathbb{P}(A)$. The model does not really talk about whether a run is in a set not in $\mathcal{F}$.

1.1.2. Random variables

If we observe a particular value $e = X(\omega)$ through a projection $X : \Omega \to E$, then we know that the run that happened lies in $X^{-1}(\{e\}) \subset \Omega$. But we're only allowed to know if ω is in $\mathcal{F}$-sets, so the only (sets of) values $A \subset E$ (including the case $A = \{e\}$) we are allowed to observe from X must have measurable preimages $X^{-1}(A) \in \mathcal{F}$.



When we observe $e = X(\omega) \in E$, we know that $A = X^{-1}(e)$ happened, but not which ω exactly.

Thus to model observables we use *measurable* functions $X : \Omega \to E$ into measure spaces E, and we call X a *random variable*.

In this sense, $\mathcal{F}$ is a restriction on the nature of all random variables, containing the information *that is allowed to be known* and revealed by any observables.

2. Experiments that happen in time

Imagine we're observing an experiment that is laid out as a process over time $\mathbb{N}$.

2.1. Atomicity of history

A single run ω of the experiment contains the whole history of the process. That is, from a mathematical point of view, the future is predetermined.

As time evolves in the real world, though, we can only observe what is *currently* happening, and not the whole history, even though in theory it already "happened". Many runs of the experiment may have the same observable properties up to a particular time $n \in \mathbb{N}$, thus we only know that the run ω we're observing lies in a (probably large) measurable subset of Ω containing runs of the same observed history up to now but different futures.

For any n, let $\mathcal{F}_n \subset \mathcal{F}$ contain all maximal sets of events that share a common observable property up to time n or earlier. That is, $\mathcal{F}_n$ contains only, and all, sets of runs that a particular run can be determined to lie in, by observing it up to time n. Then trivially $\mathcal{F}_n \subset \mathcal{F}_m$ for all $n \leq m$.

If two runs ω and ω' differ in no observable way up to time n, they are not $\mathcal{F}_n$-separable[1].

Thus in order to model slowly-appearing history over Ω (recall, Ω is a set of atoms ω containing entire individual histories) we introduce a nested sequence of σ-algebras in $\mathcal{F}$

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_n \subset \ldots$$

called a *filtration* over Ω. It is a way to *lay out in time* the otherwise indivisible experiment runs ω without actually chopping them in time pieces. So at each step n we're allowed to know only which event in $\mathcal{F}_n$ has occurred and nothing more specific (like which run ω has happened).

2.2. Random processes

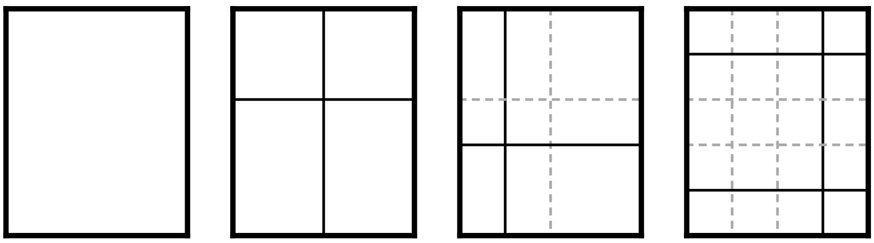
A sequence of real-world observations is then modelled by

- projections $X_n : \Omega \to E$ that distinguish only properties observable up to time n,
- i.e. $X_n(\omega) \neq X_n(\omega')$ (with both values in disjoint measurable sets in E) is only allowed if ω and ω' differ *in an observable way up to time* n, so that X_n does not accidentally reveal information about ω that is not supposed to be known at time n.
- That is, the preimages of observable (sets of) values in E must be maximal w.r.t. observables up to time n,
- i.e. $X_n^{-1}(A)$ must be in $\mathcal{F}_n$ for all measurable $A \subset E$.

Or, X_n must be $\mathcal{F}_n$-measurable for all n. Such a sequence $X_\bullet$ is called a *stochastic process*.

2.3. Filtration example diagram

Schematically we can imagine a filtration on a rectangular Ω like this:



$\mathcal{F}_0 = \{\emptyset, \Omega\}$ $\quad$ $\mathcal{F}_1$ $\quad$ $\mathcal{F}_2$ $\quad$ $\mathcal{F}_3$

A run ω is a point in the rectangle. Each of the four rectangles represents the same Ω but with a different σ-algebra on it. The σ-algebras $\mathcal{F}_n$ contain all rectangles outlined in each case (both dashed and non-dashed[2]):

1. In the beginning, we know nothing (except that the run *happened*), so no subrectangles.
2. In $\mathcal{F}_1$, the model allows to observe which of the four regions ω lies in, but nothing more.
3. In $\mathcal{F}_2$, we're allowed to observe which of the 9 (smaller!) rectangles it is in, and so on.

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[1]I borrow that term from topology. "Two points x, y are *separated* by $\mathcal{F}_n$" means that there are sets $A, B \in \mathcal{F}_n$ that contain one but not the other: $x \in A, y \notin B$ and $x \in B, y \notin A$.

[2]non-dashed borders indicate the newly introduced measurable sets that weren't measurable in the previous step

2.4. Example

For an experiment with 3 consecutive coin flips we can put

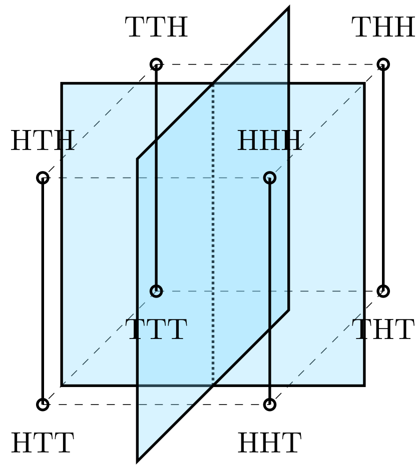
$$\Omega = \{\text{Heads}, \text{Tails}\}^3,$$

a discrete cube in 3D.

$\mathcal{F}_1$ contains only the front and back halves (apart from the whole cube and the empty set), $\mathcal{F}_2$ is generated by the four vertical edges, and $\mathcal{F}_3$ — by all 8 corners of the cube.

If $X_i : \Omega \to \{\text{Heads}, \text{Tails}\}$ is the outcome of the i th coin flip, then X_i is allowed to (though does not necessarily have to) differ only on different sets from $\mathcal{F}_i$, which it does.

Again, keep in mind that the entire runs of the experiment are fully encoded in the actual corners. The random process X_1, X_2, X_3 is only a sequence revealing smaller and smaller subsets in Ω around the true ω.



$\mathcal{F}_2$ is generated by the four vertical edges, i.e. the outcomes of the first two flips. Both X_1 and X_2 are $\mathcal{F}_2$-measurable, though neither of the two on its own separates the entire $\mathcal{F}_2$, only the vector (X_1, X_2) does.

3. Stopping times

Here we consider a property that is present up to some time point (potentially infinite) in the observed process (or equivalently, a property present only from some time point onward in the observed process).

Again, we model the time layout of the process by a filtration $\mathcal{F}_\bullet$. Then the presence of a property like above is modelled by a monotonic $\mathcal{F}_\bullet$-adapted boolean sequence $P_\bullet$ (with values in $2 = \{0, 1\}$ with the $\mathcal{P}(2)$ σ-algebra).

It is natural to ask now, what is the turning point? It is given by

$$T_X = \inf_{P_n = 1} n = \inf \{n \in \mathbb{N} : P_n = 1\}$$

At any time $n \in \mathbb{N}$, P_n tells us whether P has happened up to now. In terms of T_X, this is the event $T_X \leq n$, so it is $\mathcal{F}_n$ measurable.

Conversely, for every $\mathbb{N}$-valued random variable T with $\mathcal{F}_n$-measurable $T \leq n$ for every n, we can consider a monotone boolean sequence

$$X_n = \mathbb{1}_{\{T \leq n\}}$$

and then T is given from $X_\bullet$ as the infimum above.

A *stopping time* for a filtration $\mathcal{F}_\bullet$ is any $\mathbb{N}$-valued random variable for which

$$\forall n : \qquad \{T \leq n\} \text{ is } \mathcal{F}_n\text{-measurable},$$

i.e. it is the time at which an observed property (in the sense of a monotonic boolean sequence) "stops" holding.

4. Conditional expectation

If $\mathcal{G} \subset \mathcal{F}$ are σ-algebras on Ω, and $X : \Omega \to E$ is a function,

then	$\mathcal{G}$-measurability	is a stronger	restriction on X than $\mathcal{F}$-measurability,
because	$X^{-1}(A) \in \mathcal{G}$		condition than $X^{-1}(A) \in \mathcal{F}$.

That is, an event $A \in E$ is allowed to reveal less information on the run ω in the $\mathcal{G}$ case than in the $\mathcal{F}$ case, just because $\mathcal{G}$ contains less information than $\mathcal{F}$.

Thus an $\mathcal{F}$-measurable X need not be $\mathcal{G}$-measurable.

TODO

5. Martingales

Recall that an $\mathcal{F}_\bullet$**-martingale** is an $\mathcal{F}_\bullet$-adapted process $X_\bullet$ in L_1 with zero-$\mathcal{F}_n$-expectation increments $\Delta X_{n+1} = X_{n+1} - X_n$. By *integrable* we mean that $\mathbb{E}[|X_n|]$ exists for all n.

Definition 5.1 (Martingale): An $\mathcal{F}_\bullet$-adapted process $X_\bullet$ in $L_1(\Omega, \mathcal{F}, \mathbb{P})$ is a called a **martingale** if

$$\forall n : \mathbb{E}[\Delta X_{n+1} | \mathcal{F}_n] = 0.$$

I prefer that phrasing of the definition because I find it more natural than the standard one. By writing

$$0 = \mathbb{E}[\Delta X_{n+1} | \mathcal{F}_n] = \mathbb{E}[X_{n+1} | \mathcal{F}_n] - \mathbb{E}[X_n | \mathcal{F}_n]$$

as $\mathbb{E}[X_{n+1} | \mathcal{F}_n] = \mathbb{E}[X_n | \mathcal{F}_n]$ and applying $\mathcal{F}_n$-measurability to X_n, one gets equivalently

Definition 5.2 (Martingale, standard): An $\mathcal{F}_\bullet$-adapted process $X_\bullet$ in $L_1(\Omega, \mathcal{F}, \mathbb{P})$ is a called a **martingale** if

$$\forall n : \mathbb{E}[X_{n+1} | \mathcal{F}_n] = X_n.$$

Note that we can always decompose a process into the sum of its increments up a given time, i.e.

$$\forall n : X_n = X_0 + \sum_{i=1}^{n} \Delta X_i.$$

Here again ΔX_i denotes the "backwards" increment, $\Delta X_{i+1} = X_{i+1} - X_i$.

Now it is natural to ask, what happens if we scale the increments of a martingale, i.e. multiply each term in the above sum by some factor.

Definition 5.3 (Martingale transform): For an $\mathcal{F}_\bullet$-adapted $X_\bullet$ and a predictable $H_\bullet$, the martingale transform of $X_\bullet$ by $H_\bullet$ is denoted by $(H \blacktriangleright X)_\bullet$ and defined by

$$(H \blacktriangleright X)_n := \sum_{i=1}^{n} H_i \Delta X_i.$$

Using this notion, a martingale can be characterized not only by having its *individual* increments-means vanish, but by having (only) the final expectation of all of its predictable increments-transforms vanishes:

Proposition 5.1: Let $X_\bullet$ be $\mathcal{F}_\bullet$-adapted, with a time horizon N.

$X_\bullet$ is a martingale if and only if for any predictable $H_\bullet$, $\mathbb{E}[(H \blacktriangleright X)_N] = 0$.

6. Discrete-time market model

This section follows the first chapter of the book “Introduction to Stochastic Calculus Applied to Finance” (Lamberton & Lapeyre), whose explanations I found missing or unsatisfactory so I enhance the presentation with as much motivation as possible.

The market price vector is a stochastic process

$$S_i, \ i = 0, \dots$$

with values in $\mathbb{R}^{1+d}$ adapted to a filtration

$$\mathcal{F}_i, \ i = 0, \dots$$

of the market state known at time i. We consider $d \in \mathbb{N}$, $\mathcal{F}_\bullet$ and $S_\bullet$ fixed from now on.

The number $1 + d$ reflects that we track d risky assets and one riskless asset (the time-value of money) in the first component of $\mathbb{R}^{1+d}$.

Definition 6.1: A *trading strategy* is any $\mathbb{R}^{1+d}$-valued *predictable* sequence

$$\phi_i, \ i = 0, \dots$$

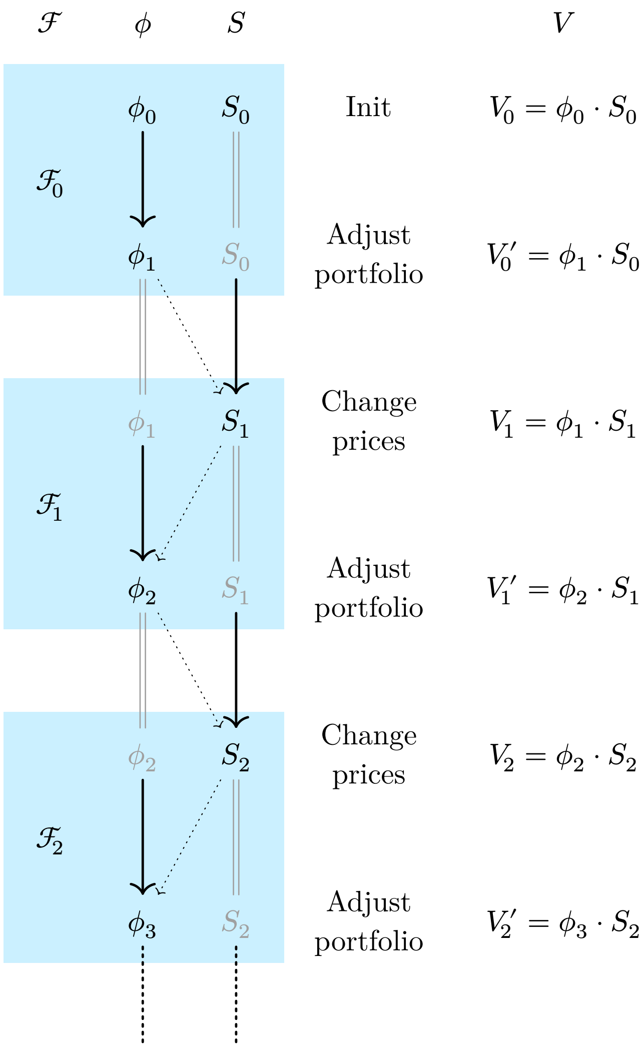
with respect to that filtration.

Predictable means that ϕ_{i+1} is $\mathcal{F}_i$-measurable, i.e. the information used (revealed) by the *next* portfolio ϕ_{i+1} is restricted to the information provided by the current market state $\mathcal{F}_i$, which is only guaranteed to contain the current stock prices (because S_i is $\mathcal{F}_i$-measurable) and past stock prices (because $\mathcal{F}_i \subset \mathcal{F}_{i+1}$).

Definition 6.2: The *value of the portfolio* defined by a strategy $\phi_\bullet$ is

$$V_i(\phi) := \phi_i \cdot S_i.$$

Now a slight confusion may arise as to what is the time order of the quantities S_i and ϕ_i. They both use the same index i so we might be under the impression that the time evolution is described by a sequence of pairs (ϕ_i, S_i). While not false, this may be deceiving when we define *self-financing* strategies. Consider the diagram on the right.



Here, in each timebox $\mathcal{F}_i$, first prices change and then the portfolio is adjusted, so we have an alternating sequence of price and portfolio changes.
The strategy ϕ is then self-financing iff the portfolio value doesn't change within a time-box ($V_i = V_i'$), but only responds to stock price changes $S_i \to S_{i+1}$.

Diagonal dotted arrows indicate the causal relationship between the time-evolution of the strategy and the stock prices in the process $(\phi_0, S_0), (\phi_1, S_1), \dots$.

Notation 6.3: Denote the *portfolio adjustment* (again, non-standard terminology) performed at time i by

$$\Delta\phi_i := \phi_{i+1} - \phi_i.$$

(consider the diagram above to convince yourself that $\Delta\phi_i$ is $\mathcal{F}_i$-measurable)

It is a vector of the sells/buys of each stock performed at time i. That is,

$\Delta\phi_i^j$ is the amount of stock j that is sold (if $\Delta\phi_i < 0$) or bought (if $\Delta\phi_i > 0$) at time i.

Notation 6.4: Denote the *price changes* that happened at time $n+1$ by

$$\Delta S_{n+1} := S_{n+1} - S_n.$$

Remark 6.5: $\Delta\phi_n$ was defined as the *forward* difference (between times $n+1$ and n), while the ΔS_n was defined as the *past* difference (between times n and $n-1$), so that both $\Delta\phi_n$ and ΔS_n are $\mathcal{F}_n$-measurable.

We also get a Leibniz rule for $V_\bullet$ $(\mathrm{d}V = \mathrm{d}\phi \cdot S + \mathrm{d}S \cdot \phi)$.

Remark 6.6:

The two increments let us express the portfolio value with a recursive formula:

$$\underbrace{V_{n+1}(\phi)}_{\text{next value}} = \underbrace{V_n(\phi)}_{\text{prev value}} + \underbrace{\Delta\phi_n \cdot S_n}_{\text{changes in strategy}} + \underbrace{\phi_{n+1} \cdot \Delta S_n}_{\text{changes in prices}}.$$

That is, the next value is the previous value plus the changes from the strategy and the changes in the prices.

6.1. Self-financing strategies

When the second term in the sum on the right in the remark above, $\Delta\phi_n \cdot S_n$, vanishes for all n, we get a very important type of strategies. Denote the first part of recursive formula by

$$V_n'(\phi) = V_n(\phi) + \Delta\phi_n \cdot S_n,$$

so that

$$V_{n+1}(\phi) = V_n'(\phi) + \phi_{n+1} \cdot \Delta S_n.$$

Those two mutually-recursive formulas reflect the structure of the alternating sequence $V_0, V_0', V_1, V_1', \ldots$ in the diagram above, so

Remark 6.1.1: $V_n'(\phi)$ is the *adjusted value* of the portfolio **after** the next strategy choices ϕ_{n+1} are applied but **before** the prices have been updated.

Now the condition that the second term in the earlier recursive formula for $V(\phi)$ vanishes for all n can be phrased simply as the equality $V(\phi) = V'(\phi)$ (as processes). It is instructive to derive a direct expression for V':

$$\begin{aligned} V_n'(\phi) &\equiv V_n(\phi) \qquad + \Delta\phi_n \cdot S_n, \\ &= \cancel{\phi_n \cdot S_n} \quad + (\phi_{n+1} \cdot S_n - \cancel{\phi_n \cdot S_n}) \\ &= \phi_{n+1} \cdot S_n. \end{aligned}$$

Definition 6.1.2: A strategy ϕ_i is *self-financing* if the value of the portfolio stays the same after the strategy adjustment:

$$V_i'(\phi) = V_i(\phi) \qquad \text{for all } i,$$

i.e.

$$\phi_{i+1} \cdot S_i = \phi_i \cdot S_i \qquad \text{for all } i.$$

Now we can get back to the condition on the vanishing second term in the recursive formula for V:

Remark 6.1.3: The self-financing condition $V_i = V_i'$ is equivalent to

$$\Delta\phi_i \cdot S_i = 0,$$

i.e. all buys and sells cancel each other in value, i.e. no money is lost or needs to be brought in for the adjustment.
Then a self-financing strategy is one where the recursive formula for V has a vanishing second term, i.e.

$$V_{n+1}(\phi) = V_n(\phi) + \phi_{n+1} \cdot \Delta S_n.$$

We can apply the last formula iteratively and write

$$V_n(\phi) = V_0(\phi) + \sum_{i=1}^{n} \phi_n \cdot \Delta S_n$$

Note how close this expression is to Definition 5.3 — we could try to see it as a martingale transform under some conditions (though, the quantities ϕ_n and S_n are vectors, so it would rather be a *sum* of the component-wise martingale transforms). However, even if $V_0(\phi)$ were to vanish, ΔS_n would generally not be a martingale because the riskless asset price process $S^0_\bullet$ increases (maybe even deterministically). Instead, we can consider the discounted price process $\tilde{S}_\bullet$ and the corresponding discounted portfolio value $\tilde{V}_\bullet$.

Remark 6.1.4: For a self-financing strategy with $\tilde{V}_0(\phi) = 0$, if each $\tilde{S}^j_\bullet, j = 0, \ldots, d$ were a martingale, then $\tilde{V}_\bullet(\phi)$ would be a sum of the martingale transforms of the prices by the strategy,

$$\tilde{V}_\bullet(\phi) = \sum\nolimits_j (\phi^j \blacktriangleright \tilde{S}^j)_\bullet.$$

We'll get back to this observation in the next section.

It should be close to mind that even for a strategy that is not self-financing (i.e. some of the $\Delta\phi_n \cdot S_n$ does not vanish), if we were to put the quantity $\Delta\phi_n \cdot S_n$ into ϕ^0_n, i.e. if we invest (borrow) the surplus (shortage) of sells-minus-buys into (from) the riskless asset, we'd get a self-financing strategy.

Proposition 6.1.1: Any $\mathbb{R}^d$-valued predictable sequence (a "strategy" only on the risky assets)

$$\left(\phi^1_n, \ldots, \phi^d_n\right) \qquad \text{for} \quad n = 0, 1, \ldots$$

is a restriction of a unique self-financing strategy ($\mathbb{R}^{1+d}$-valued)

$$\left(\boldsymbol{\phi^0_n}, \phi^1_n, \ldots, \phi^d_n\right) \qquad \text{for} \quad n = 0, 1, \ldots$$

for any choice of an initial value $V_0(\phi) \in \mathbb{R}$ (or a choice of any of ϕ^0_n for $n \geq 0$).

In other words, the self-financing strategies restricting to a given $\left(\phi^1_n, \ldots, \phi^d_n\right)_n$ are a one-parameter family indexed by V_0.

Proof: Fix $\underline{\phi}_n = \left(\phi^1_n, \ldots, \phi^d_n\right)$ for all $n \geq 0$. Any strategy $\phi_\bullet$ that (on the risky assets) restricts to this process is determined by the choices of bank account amounts $\varphi^0_\bullet$.

The self-financing condition $\phi_n \cdot S_n = \phi_{n+1} \cdot S_n$ imposes a restriction on $\varphi^0_\bullet$, though, and reads

$$\underbrace{\left(\phi^0_{n+1} - \phi^0_n\right)} \cdot S^0_n + \sum_{i=1}^{d} \left(\phi^i_{n+1} - \phi^i_n\right) \cdot S^i_n = 0 \qquad \text{for all } n = 0, 1, \ldots,$$

or

$$\phi^0_{n+1} = \frac{1}{S^0_n} \left(\phi^0_n \cdot S^0_n - \sum_{i=1}^{d} \left(\phi^i_{n+1} - \phi^i_n\right) \cdot S^i_n \right) \qquad \text{for all } n = 0, 1, \ldots.$$

Here a choice of any element of the sequence $\phi^0_\bullet$ determines the rest, so the self-financing strategies restricting to $\underline{\phi}_n$ are a one-parametric family indexed by (e.g.) ϕ^0_0.

Since $V_0 \equiv \phi^0_0 \cdot S_0 + \sum_{i=1}^{d} \phi^i_0 \cdot S^i_0$ establishes a 1:1 relation between V_0 and ϕ^0_0 (given $\underline{\phi}_0$), we can also consider this parameter to be V_0. □

6.2. Linear algebra

Remark 6.2.1: The set of strategies on Ω clearly forms a vector space under pointwise addition and scalar-multiplication, denote it $\mathsf{Strategy}$.

With a finite time horizon N, $\mathsf{Strategy} = \mathbb{R}^{(1+d)\times(1+N)\times\Omega}$,
and with an infinite horizon $\mathsf{Strategy} = \mathbb{R}^{(1+d)\times\mathbb{N}\times\Omega}$.

Self-financing strategies are filtered from it by the linear condition $\phi_{n+1}S_n = \phi_n S_n$, so they form a subspace $\mathsf{SelfFinancing} \subset \mathsf{Strategy}$.

Real random variables on Ω form a vector space too, denote it by $\mathbb{R}^\Omega$.

For each n and each strategy ϕ, $V_n(\phi)$ is a random variable, so a vector $V_n(\phi) \in \mathbb{R}^\Omega$. Now we can consider how the dependence on ϕ looks like:

Remark 6.2.2: For any n, the operator $V_n(\phi) = \phi_n \cdot S_n$ is linear in ϕ. Its signature is

$$V_n : \mathsf{Strategy} \to \mathbb{R}^\Omega$$

Then the images of $\mathsf{Strategy}$ and $\mathsf{SelfFinancing}$ are linear subspaces of $\mathbb{R}^\Omega$:

$$V_n(\mathsf{SelfFinancing}) \subset V_n(\mathsf{Strategy}) \subset \mathbb{R}^\Omega.$$

(the last inclusion might actually be an equality, TODO check)

We will talk about strategies of zero initial capital, so it is useful to have

Notation 6.2.3: Denote the subsets of the above strategy spaces of those strategies that satisfy $V_0(\phi) = 0$ by a superscript 0: $\mathsf{SelfFinancing}^0, \mathsf{Strategy}^0$.
Clearly they form subspaces of the respective spaces.

These terms allow us to reformulate Proposition 6.1.1 in a clearer way. For any $\phi = (\phi^0, \phi^1, ..., \phi^d) \in \mathsf{Strategy}$, denote by $\phi_\flat$ the risky part of ϕ, $\phi_\flat := (\phi^1, ..., \phi^d)$. Then the projection

$$\mathsf{Strategy}_{1+d} \to \mathsf{Strategy}_d$$
$$\phi \mapsto \phi_\flat$$

is a linear map, where the d and $d+1$ subscripts denote the number of assets the space considers.

Remark 6.2.4: Proposition 6.1.1 establishes that the linear map

$$\mathsf{SelfFinancing}_{1+d} \cong \mathsf{Strategy}_d \times \mathbb{R}$$
$$\phi \mapsto (\phi_\flat, V_0(\phi))$$

is in fact an isomorphism.

Remark 6.2.5: This allows us to compute $\dim(\mathsf{SelfFinancing}_d) = \dim\big(\mathbb{R}^{d\times(1+N)\times\Omega} \times \mathbb{R}\big) = 1 + d \times (1+N) \times \Omega$ and so

$$\begin{aligned}\operatorname{codim}(\mathsf{SelfFinancing}_{1+d}) &= \dim(\mathsf{Strategy}_d) \qquad\qquad - \dim(\mathsf{SelfFinancing}_{1+d}) \\ &= (1+d)\times(1+N)\times\Omega - (d\times(1+N)\times\Omega+1) \\ &= (1+N)\times\Omega - 1.\end{aligned}$$

6.3. Arbitrage

Definition 6.3.1 (Admissible strategy): A *self-financing* strategy is called **admissible** if $V_n(\phi) \geq 0$ for all n (and all $\omega \in \Omega$).

Remark 6.3.2: The set of admissible strategies is closed under addition and non-negative scalar multiplication, so it forms a cone $\mathsf{Admissible} \subset \mathsf{SelfFinancing}$.

Its image under V_n is then a cone $V_n(\mathsf{Admissible}) \subset V_n(\mathsf{SelfFinancing}) \cap \mathbb{R}^{\Omega}_{+}$.

Like above, denote the set of zero initial capital admissible strategies by $\mathsf{Admissible}^0$.

Thus we get the following chain of strategy classes (here N is either a finite number or equals $\mathbb{N}$, and $\leq$ means $\subset$ but for subspaces):

$$\begin{array}{ccccccccc} & \supset & \mathsf{Admissible} & \leq & \mathsf{SelfFinancing} & \leq & \mathsf{Strategy} & \leq & \\ \mathbb{R}^{(1+N)\times(1+d)\times\Omega}_{\geq 0} & & \cup & & \geq & & \geq & & \mathbb{R}^{(1+N)\times(1+d)\times\Omega}, \\ & \supset & \mathsf{Admissible}^0 & \leq & \mathsf{SelfFinancing}^0 & \leq & \mathsf{Strategy}^0 & \leq & \end{array}$$

and for each $n \leq N$, the corresponding chain of images under $V_n : \mathbb{R}^{N\times\Omega} \to \mathbb{R}^{\Omega}$:

$$\begin{array}{ccccccccc} & \supset & V_n(\mathsf{Admissible}) & \leq & V_n(\mathsf{SelfFinancing}) & \leq & V_n(\mathsf{Strategy}) & \leq & \\ \mathbb{R}^{\Omega}_{\geq 0} & & \cup & & \leq & & \geq & & \mathbb{R}^{\Omega}. \\ & \supset & V_n(\mathsf{Admissible}^0) & \leq & V_n(\mathsf{SelfFinancing}^0) & \leq & V_n(\mathsf{Strategy}^0) & \leq & \end{array}$$

Horizon: From here on, we fix an $N \in \mathbb{N}$, called the (time) *horizon*.

Remark 6.3.3: All indices n, i below are reserved for time and vary between 0 and N.

j is reserved for assets and varies between 0 and $1 + d$.

The market is said to have an **arbitrage opportunity** if some admissible strategy with zero initial value delivers a strictly positive value on a non-null set.
That is, an arbitrage opportunity is a

zero-investment	**risk-free**	**chance of profit**.
($V_0 = 0$)	($V_n \geq 0$ for *every* ω and n)	($V_N \gneqq 0$ on a non-null set)

A market without arbitrage opportunities is called viable.

Definition 6.3.4 (Viable/arbitrage-free market): The market is called **arbitrage-free** or **viable** if every admissible strategy with $V_0(\phi) = 0$ satisfies $V_N(\phi) = 0$.

In terms of the notation above, a market is viable if $V_N(\mathsf{Admissible}^0) = 0$.

Spelled out fully, the market is viable if for all self-financing ϕ,

$$\text{if} \quad \begin{matrix} \forall n : V_n(\phi) \geq 0 \\ \text{and } V_0(\phi) = 0 \end{matrix}, \text{ then } V_N(\phi) = 0.$$

As a silly (counter)example, if

$$\begin{aligned} &\text{at} \quad n = 0, \quad S_0^1 = S_0^0 \\ &\text{at} \quad n > 0, \quad S_n^1 = 2 \cdot S_n^0, \end{aligned}$$

i.e. the price of risky asset 1 equals the riskless at first but then becomes twice as much as the riskless one, then the constant strategy $\phi_n = (-1, 1)$ has zero initial value and $V_n = -S_n^0 + S_n^1 = S_n^0 > 0$. Thus the market defined by those $S_\bullet^\bullet$ is not viable.

A viable market was defined by a condition on the terminal portfolio value $V_N(\phi) = 0$ for all *admissible* strategies, but it is equivalent to require it from all self-financing strategies:

Proposition 6.3.1: The risk-freeness condition ($V_n(\phi) \geq 0$) for arbitrage-free markets can also be required only for horizon time N:

A market is viable	if and only if	no strategy $\phi \in \mathsf{SelfFinancing}^0$ can provide $V_N(\phi) \in \mathbb{R}_{\geq 0}^{\Omega} \setminus \{0\}$.

Proof: The ($\Leftarrow$) direction is trivial, just because $\mathsf{Admissible}^0 \subset \mathsf{SelfFinancing}^0$.

For ($\Rightarrow$), assume that no $\phi \in \mathsf{Admissible}^0$ gives $V_N(\phi) \in \mathbb{R}_{\geq 0}^{\Omega} \setminus \{0\}$ and take a $\phi \in \mathsf{SelfFinancing}^0$ to show $V_N(\phi) \notin \mathbb{R}_{\geq 0}^{\Omega} \setminus \{0\}$.

If $\phi \in \mathsf{Admissible}^0$, we're done; now assume ϕ is not admissible, i.e. gives a negative portfolio value $V_n(\phi) < 0$ with positive probability at some time n.

If $n = N$, we're done; now take n to be the last such one. Then the strategy ϕ with positive probability has negative value at $n < N$ and at all later times gives $V_\bullet(\phi) \in \mathbb{R}_{\geq 0}^{\Omega}$.

TODO □

6.3.1. Fundamental theorem of asset pricing

Proposition 6.3.1.1 (Fundamental theorem of asset pricing):

A market is arbitrage-free if and only if under some equivalent measure, the discounted price processes $\tilde{S}^k_\bullet$ are martingales, $k = 0, 1, \ldots$.

Before we present the proof, this theorem deserves a bit of discussion.

Now the L&L book proceeds to prove this directly. I find the presentation of the proof there far from pedagogical, because they combined 4 distinct concepts in the same proof without clear indication which one is used where and how they are connected:

1. Interpreting viability in terms of only at-the-end lack of risk;
2. Treating martingales as processes having all their increments-transforms of zero-mean;
3. Relationship between value of self-financing strategies and martingale transforms;
4. Separation of a convex set from a linear subspace;

They use all (1),(2),(4) only in the ($\Rightarrow$) direction, and (3) only for ($\Leftarrow$), and to me that makes the easy direction's (the one that does not use convex separation) proof look mysterious. But if we notice that (1) and (2) concern exclusively the left and right side of the equivalence, respectively, and (3) applies to each side individually, we can apply them upfront and make the proof conceptually shorter.

(1) and (2) were already discussed as propositions above, so we can readily install them in the two sides of the theorem statement.

Applying the equivalence from Proposition 6.3.1 and the characterization of martingales in Proposition 5.1, the proposition reads

For any $\phi \in \mathsf{SelfFinancing}^0$, if $V_N(\phi) \geq 0$, then $V_N(\phi) = 0$.

if and only if

there exists $\mathbb{P}^*$ equivalent to $\mathbb{P}$, such that for any predictable $H_\bullet$,

$$\mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0 \quad \text{for all } j.$$

Remark 6.3.1.1: I find this phrasing illuminating, because by Remark 6.1.4 we know that self-financing strategies of zero initial value, when applied on martingale prices, act like martingale transforms. And here on the left side we have exactly such strategies, and on the right side - martingale transforms of the prices.

In fact, can we rewrite the right side in terms of portfolio values of zero-investment self-financing strategies then? That is, can we say that

Proposition 6.3.1.2: The way predictable $\mathbb{R}$-valued sequences act on process increments and the portfolio value of self-financing strategies with zero initial capital are related by the equivalence

For all predictable $\overset{(\mathbb{R}\text{-valued})}{H}$ and each $\overset{(\text{asset})}{j}$,	if and only if	for all $\phi \in \mathsf{SelfFinancing}^0$,
$\mathbb{E}\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$		$\mathbb{E}\left[\tilde{V}_N(\phi)\right] = 0$

(notice the similarity between the first lines on each side as well as between the second lines)

Proof: As discussed in Remark 6.1.4, for a self-financing strategy, the discounted portfolio value is a martingale transform of the market prices, $\tilde{V}_N(\phi) = \sum_{j=0}^{d} \left(\phi^j \blacktriangleright \tilde{S}^j\right)_N$, so by linearity of $\mathbb{E}$ the right side condition can be replaced by $\sum_j \mathbb{E}\left[\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right] = 0$. We prove the rest in the next proposition. □

Proposition 6.3.1.3:

For all predictable $\overset{(\mathbb{R}\text{-valued})}{H}$ and each $\overset{(\text{asset})}{j}$,	if and only if	for all $\phi \in \mathsf{SelfFinancing}^0$,
$\mathbb{E}\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$		$\sum_j \mathbb{E}\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0$

Proof (⇒): Assume the left side and take a $\phi \in \mathsf{SelfFinancing}^0$. All ϕ^j are surely predictable, so each $\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N$ has zero mean and the total gain expectation vanishes:

$$\mathbb{E}^*\left[\tilde{V}_N\right] = \mathbb{E}^*\left[\sum_j \left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right] = \sum_j \underbrace{\mathbb{E}^*\left[\left(\phi^j \blacktriangleright \tilde{S}^j\right)_N\right]}_{0} = 0.$$

□

Before showing the other direction, a brief comment:

Remark 6.3.1.2: Notice that $\tilde{S}^0$ is constant, so trivially $H \blacktriangleright \tilde{S}^0 = 0$ and the condition on the left side here matters only for $j > 0$. This is why in the proof in the book, instead of self-financing strategies, they consider predictable processes in $\mathbb{R}^d$ (and not $\mathbb{R}^{d+1}$). But I find this unnecessarily confusing, as we can reuse terms already established ($\mathsf{Admissible}^0$) and just point out that the j on the right hand side can vary from 1 onwards.

Proof (⇐): Take a predictable H and an asset j, we have to prove that $\mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0$.

If $j = 0$, we are done by the preceding remark, so now H can be regarded as a strategy operating only on a risky asset $j > 0$.
By Proposition 6.1.1 we can extend H to a (unique) self-financing strategy ϕ with $V_0(\phi) = 0$.

But for $\phi \in \mathsf{SelfFinancing}^0$ we've assumed

$$\sum_j \mathbb{E}\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0.$$

All sum terms except $i = j$ vanish because for $i = 0$, $\tilde{S}^i$ is constant; while for $i > 0$ and $i \neq j$, ϕ^j is zero. And ϕ^j is just H, so the sum consists simply of the term $\mathbb{E}(H \blacktriangleright S^j)_N$ and it is zero.□

Now we can rephrase the right side of the fundamental theorem of asset pricing again:

For any $\phi \in \mathsf{SelfFinancing}^0$, if $V_N(\phi) \geq 0$, then $V_N(\phi) = 0$.	if and only if	there exists $\mathbb{P}^*$ equivalent to $\mathbb{P}$, s.t. for any $\phi \in \mathsf{SelfFinancing}^0$, $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right] = 0.$

Assuming that $\mathcal{F} = \mathcal{P}(\Omega)$ and $\mathbb{P}(\omega) > 0$ for all outcomes $\omega \in \Omega$, the equivalence $\mathbb{P}^* \sim \mathbb{P}$ means $\forall \omega : \mathbb{P}^*(\omega) > 0$ as well. A mere geometrical observation should now make the statement almost obvious:

Remark 6.3.1.3: Considering the object $\mathbb{P}^*$ as a vector in $\mathbb{R}^\Omega$ (only works when $\mathcal{F} = \mathcal{P}(\Omega)$), for any ϕ the mean $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right]$ is just the dot product $\mathbb{P}^* \cdot \tilde{V}_N(\phi)$. Then the condition $\mathbb{E}^*\left[\tilde{V}_N(\phi)\right] = 0$ for all $\phi \in \mathsf{SelfFinancing}^0$ means just that $\mathbb{P}^*$ must be orthogonal to $\tilde{V}_N(\mathsf{SelfFinancing}^0)$.

So the theorem statement can be equivalently phrased as

Proposition 6.3.1.4 (Fundamental theorem of asset pricing, again):

$$V_N(\mathsf{SelfFinancing}^0) \cap \mathbb{R}^\Omega_{\geq 0} = \{0\} \quad \text{if and only if} \quad \exists \mathbb{P}^* \in \mathbb{R}^\Omega_{\geq 0} : \mathbb{P}^* \perp \tilde{V}_N(\mathsf{SelfFinancing}^0)$$

Now *that* phrasing should be geometrically quite clear.

Proof:

($\Leftarrow$). Assume $\mathbb{P}^*$ equivalent to $\mathbb{P}$ such that

$$\mathbb{P}^* \perp \tilde{V}_N(\mathsf{SelfFinancing}^0).$$

Take an element

$$f \in \underbrace{V_N(\mathsf{SelfFinancing}^0)} \cap \underbrace{\mathbb{R}^\Omega_{\geq 0}}$$

$$\text{Then} \qquad \mathbb{P}^* \perp f \quad \text{and} \quad f \geq 0,$$

i.e. $\mathbb{E}^*(f) = 0$ and $f \geq 0$, so $f \overset{\mathbb{P}^*}{=} 0$.

($\Rightarrow$). Assume

$$\tilde{V}_N(\mathsf{SelfFinancing}^0) \cap \mathbb{R}^\Omega_{\geq 0} = \{0\}.$$

The left term is a linear space and the right term is convex, so by a standard separation theorem there exists a separating functional $\mathbb{E}^*$ on $\mathbb{R}^\Omega$ vanishing on $\tilde{V}_N(\mathsf{SelfFinancing}^0)$ and positive on $\mathbb{R}^\Omega_{\geq 0}$, or — a unit (by $\|\cdot\|_1$) vector $\mathbb{P}^* \in \mathbb{R}^\Omega_{\geq 0}$ orthogonal to $\tilde{V}_N(\mathsf{SelfFinancing}^0)$.

□

6.3.2. Summary

We summarize the proof and the preceding rewrites of the Fundamental theorem of asset pricing (FTAP).

Proposition 6.3.2.1 (Fundamental theorem of asset pricing): For a market defined by the $\mathbb{R}^{1+d}$-valued process $S_\bullet$, the following equivalences hold:

$$
\begin{tikzcd}[column sep=large]
\begin{array}{c}\text{The market is}\\ \text{arbitrage-free}\end{array} \arrow[r, Leftrightarrow, "\text{FTAP}"] \arrow[dd, Leftrightarrow, "\text{By definition}"'] & \begin{array}{c}\text{Under some } \mathbb{P}^* \text{ equivalent to } \mathbb{P},\\ \text{all } \tilde{S}^j_\bullet \text{ are martingales.}\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 5.1}"] \\
 & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall H \text{ predictable } \forall j\\ \mathbb{E}^*\left[\left(H \blacktriangleright \tilde{S}^j\right)_N\right] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 6.3.1.2}"] \\
\begin{array}{c}\forall \phi \in \mathsf{Admissible}^0\\ V_N(\phi) = 0\end{array} \arrow[dd, Leftrightarrow, "\text{Proposition 6.3.1}"'] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0 :\\ \sum_j \mathbb{E}^*\left[\phi^j \blacktriangleright \tilde{S}^j\right] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Proposition 6.3.1.3}"] \\
 & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0\\ \mathbb{E}^*[V_N(\phi)] = 0\end{array} \arrow[d, Leftrightarrow, "\text{Remark 6.3.1.3}"] \\
\begin{array}{c}\forall \phi \in \mathsf{SelfFinancing}^0\\ V_N(\phi) \geq 0 \Rightarrow V_N = 0\end{array} \arrow[d, Leftrightarrow, "\text{Syntactical}"'] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} : \forall \phi \in \mathsf{SelfFinancing}^0\\ \mathbb{P}^* \cdot V_N(\phi) = 0\end{array} \arrow[d, Leftrightarrow, "\text{Syntactical}"] \\
\begin{array}{c}V_N\left(\mathsf{SelfFinancing}^0\right) \cap \mathbb{R}^\Omega_{\geq 0}\\ = \{0\}\end{array} \arrow[r, Rightarrow, shift left, "\text{Hahn-Banach}"] & \begin{array}{c}\exists \mathbb{P}^* \sim \mathbb{P} :\\ \mathbb{P}^* \perp V_N\left(\mathsf{SelfFinancing}^0\right)\end{array} \arrow[l, Rightarrow, shift left, "\text{Measure-theoretic}"]
\end{tikzcd}
$$

The proof of the bottom equivalence was given last, the proofs of the rest of the equivalences are in the cited propositions.

6.3.3. Example for $|\Omega| = 2$

The Fundamental theorem of asset pricing is essentially a statement about the space $\mathbb{R}^\Omega$, in which portfolio terminal values lie. $\mathbb{R}^\Omega$ is the space of all possible random choices for a random real number.

That space is $|\Omega|$-dimensional, so we can visualise graphically it when the probability space is as simple as $\Omega = \{\omega_1, \omega_2\}$, i.e. there are only two possible outcomes.

An arbitrage strategy will have its portfolio value lie in the first quadrant of $\mathbb{R}^{\{\omega_1, \omega_2\}}$ (that is, $\{(x, y) : x \geq 0, y \geq 0\}$), so that in no case of the two outcomes it induces a loss.

Equivalently, a no-arbitrage self-financing strategy of zero-investment will have its final portfolio value *outside* of that quadrant, so for a viable market all terminal values of $\mathsf{SelfFinancing}^0$ form a line through the origin that does not enter the first quadrant. Then that line induces a unit normal lying in that quadrant so gives positive probabilities to both outcomes (i.e. is an equivalent probability measure).

6.4. Perfect hedging (attainable claims)

For short now we'll refer to the space of self-financing strategies by SF, and those of no initial value — by SF^0.

A market was characterized as viable by its discounted asset prices being martingales under a suitable measure, a condition on the configuration of the subspace $\tilde{V}_N(\mathsf{SF}^0)$ in $\mathbb{R}^\Omega$.

Now we look at the question of uniqueness of $\mathbb{P}^*$. $\mathbb{P}^*$ was defined as a unit normal vector to $\tilde{V}_N(\mathsf{SF}^0)$ lying in $\mathbb{R}^\Omega_{\geq 0}$. A unit normal vector is unique if the orthogonal complement of $\tilde{V}_N(\mathsf{SF}^0)$ is one-dimensional, i.e. if $\tilde{V}_N(\mathsf{SF}^0)$ is of codimension 1 in $\mathbb{R}^\Omega$. Geometrically it might be clear that the converse is also true, and we will show this equivalence in a bit.

As we can expect, SF^0 is already of codimension 1 in SF (if we imagine that we let the superscript 0 run through all possible initial values), so $\tilde{V}_N(\mathsf{SF}^0)$ is of codimension at most one in $\tilde{V}_N(\mathsf{SF})$, and we can see that it is in fact *exactly* 1:

$$\mathsf{SF}^0 \underset{\text{codim}=1}{\subset} \mathsf{SF}$$
$$\tilde{V}_N(\mathsf{SF}^0) \underset{\text{codim}=1}{\subset} \tilde{V}_N(\mathsf{SF})$$

Proposition 6.4.1:

$$\dim(\mathsf{SF}) = \dim(\mathsf{SF}^0) + 1$$
$$\dim\big(\tilde{V}_N(\mathsf{SF})\big) = \dim\big(\tilde{V}_N(\mathsf{SF}^0)\big) + 1.$$

Proof: Any $\phi \in \mathsf{SF}$ decomposes it into

$$\phi = (\phi - \phi_b) + \phi_b,$$

where the strategy ϕ_b is the pure-bond strategy defined by the time-constants $\phi^0_n = V_0(\phi)$ and $\phi^j_n = 0$ for all other $j > 0$. Thus SF decomposes into $\mathsf{SF}^0 \oplus B$ where B is the space of self-financing strategies not operating on the risky assets,

$$B = \left\{\phi \in \mathsf{SF} : \phi^j_0 = 0 \text{ for all } j > 0\right\}$$

and is one-dimensional because each $\phi \in B$ is uniquely determined by its initial value ϕ^0_0.

Moreover, for $\phi \in B$ we get $\tilde{V}_N(\phi) = \phi_N \cdot \tilde{S}_N = \phi^0_N \tilde{S}^0_N = \phi^0_0 \tilde{S}^0_N$, so

$$\tilde{V}_N(B) \equiv \left\{\tilde{V}_N(\phi) : \phi \in B\right\} \overset{!}{=} \left\{\phi^0_0 \tilde{S}^0_N : \phi \in B\right\} = \left\{\alpha \tilde{S}^0_N : \alpha \in \mathbb{R}\right\},$$

i.e. $\dim\big(\tilde{V}_N(B)\big) = 1$ and $\dim\big(\tilde{V}_N(\mathsf{SF})\big) = \dim\big(\tilde{V}_N(\mathsf{SF}^0)\big) + 1$ follows. □

Remark 6.4.1: Above we noted that if $\operatorname{codim}\big(\tilde{V}_N(\mathsf{SF}^0); \mathbb{R}^\Omega\big) = 1$, then $\mathbb{P}^*$ is unique (if exists). Now we showed that $\operatorname{codim}\big(\tilde{V}_N(\mathsf{SF}^0); \tilde{V}_N(\mathsf{SF})\big) = 1$, thus the uniqueness condition can be restated as: if $\tilde{V}_N(\mathsf{SF}) = \mathbb{R}^\Omega$, then $\mathbb{P}^*$ is unique (if exists).

It turns out, the converse is true as well, which should also be kinda geometrically clear.

Proposition 6.4.2: Assume the market is viable and $\mathbb{P}^*$ is a probability under which discounted asset prices are martingales. Then,

$$\mathbb{P}^* \text{ is unique} \qquad \text{if and only if} \qquad \tilde{V}_N(\mathsf{SF}) = \mathbb{R}^\Omega.$$

Remark 6.4.2: Now *that* is a solid motivation for the definition of a *complete* market that follows.

In the L&L book market completeness is defined and the statement above is proved directly, but little insight is given about these concepts. I believe the geometrical discussion above does a much better job at motivating them and also shows how it is natural to expect that statement to be true and thus to define a complete market in that precise way.

Geometrically we can say even more, and actually *quantify* the (non-)uniqueness of $\mathbb{P}^*$.

Remark 6.4.3: Assume the market is viable and $\mathbb{P}^*$ is a probability under which discounted asset prices are martingales. Then, the set

$$E = \left\{ \mu \in \mathbb{R}^\Omega : \text{under } \mu, \tilde{S}^j_\bullet \text{ is a martingale for all } j \right\}$$

of (not necessarily normalized/equivalent) measures under which asset prices are martingales, has a dimension of

$$\dim E = \operatorname{codim}\left(\tilde{V}_N(\mathsf{SF}^0), \mathbb{R}^\Omega\right).$$

Or, after normalizing, the degrees of freedom of $\mathbb{P}^*$ equal

$$\operatorname{codim}\left(\tilde{V}_N(\mathsf{SF}^0), \mathbb{R}^\Omega\right) - 1.$$

A **contingent claim** is a promise to pay some amount h at a *maturity time* T, the amount depending on the market state. We model it by a non-negative real random variable h that is only $\mathcal{F}_T$-measurable, so the amount might not be known in advance.

Remark 6.4.4: Consider a contingent claim h at a maturity time T.

If all admissible strategies ϕ induce a portfolio value at T that is
- *less* than h, the contingent claim is essentially "free money" for the receiver;
- *greater* than h,

Definition 6.4.5: An $\mathbb{R}$-valued random variable h is called **attainable** if there exists an admissible strategy ϕ that results in the end in a portfolio of value exactly h:

$$V_N(\phi) = h.$$

Remark 6.4.6: If h is a constant $h(\omega) = M \in \mathbb{R}_+$, it is trivially attainable by the strategy with $\phi_N^0 = M$ that is the constant zero on the risky assets (exists uniquely by the proposition above, and is obviously admissible).

Even if h is only *bounded* by a constant, $h \leq M$, we could attain the (higher) value M by the same approach. Thus attainability is *not* essentially about reaching *at least* some value — we are allowed to start with as much cash in advance as we need.

In the case of a finite Ω (which is assumed in the book), h is bounded anyway. Attaining h, and not just M, is more difficult. If h is not $\mathcal{F}_0$-measurable, we cannot just prepare the cash in advance, because we are not allowed (by $\mathcal{F}_0$) to know exactly how much we will need (even if know it won't exceed M).

For example, if $\mathcal{F}_0 = \{\varnothing, \Omega\}$, the initial cash ϕ_0^0 is only allowed to be constant (by $\mathcal{F}_0$-measurability). Thus even in the bounded case we have to make use of the risky assets to achieve the precise target h, and we have to abide by $\mathcal{F}_n$-measurability all the way $n = 0, ..., N$.

Recall Remark 6.1.4 where we noted that the discounted portfolio value is a martingale transform of the discounted asset prices, $\tilde{V}_\bullet = \sum_j \phi \blacktriangleright \tilde{S}_\bullet^j$, so if $\tilde{S}_\bullet^j$ is a martingale, so is $\tilde{V}_\bullet(\phi)$ for any self-financing ϕ. But the martingale property means that $\tilde{V}_n(\phi) = \mathbb{E}^*\left[\tilde{V}_N(\phi)|\mathcal{F}_n\right]$, so

Remark 6.4.7: If h is an attainable contingent claim in a viable market, its value *now* ("now" being any time up to the horizon, $n \leq N$) is the corresponding value of a replicating strategy,

$$\begin{aligned} \tilde{V}_n(\phi) &= \mathbb{E}^*[\quad \tilde{V}_N(\phi)|\mathcal{F}_n] \\ \beta_n V_n(\phi) &= \mathbb{E}^*[\beta_N \underbrace{V_N(\phi)}_{h}|\mathcal{F}_n], \end{aligned}$$

or

$$V_n(\phi) = \frac{\beta_N}{\beta_n}\mathbb{E}^*[h|\mathcal{F}_n].$$

Definition 6.4.8: The *fair price* at time $n \leq N$ of a contingent claim h in a no-arbitrage market is the value of any replicating strategy at that time, and is denoted by π_n^h.

The remark above proves

Proposition 6.4.3: Under no arbitrage, for any attainable claim $h \geq 0$,

$$\pi_n^h = \frac{\beta_N}{\beta_n}\mathbb{E}^*[h|\mathcal{F}_n].$$

In particular, if the no-arbitrage market is deterministic at time zero ($\mathcal{F}_0 = \{\varnothing, \Omega\}$),

Remark 6.4.9: The value of h at time zero is its $\mathbb{P}^*$-weighted mean,

$$\pi_0^h = \mathbb{E}^*[h] \equiv \mathbb{P}^* \cdot h.$$

I find this quite *remark*able: in the first example of options I saw in “A course on Financial calculus”, the authors showed how pricing the option by the average of the gains gives an unfair price; but here we see that the fair price is found by properly averaging it (via $\mathbb{P}^*$).

Definition 6.4.10: The market is called *complete* if every non-negative h is attainable.

Proposition 6.4.4: Assume the market is viable. Then,

the market is complete if and only if under a **unique** $\mathbb{P}^*$ equivalent to $\mathbb{P}$, $\tilde{S}^j_\bullet$ is a martingale for all j.

7. Cox-Rox-Rubinstein model

In this model, the market consists of a single stock and a single bond, and the stock price evolution is described by two multiplicative constants u, d, *up* and *down*:

$$S_{n+1} = \begin{cases} uS_n, \text{or} \\ dS_n \end{cases} \quad \text{for all } n.$$

The main object here is the ratio

$$T_n := \frac{S_{n+1}}{S_n}.$$

Proposition 7.1: The market in this model is viable if and only if

$$\mathbb{E}^*[T_n \mid \mathcal{F}_n] = 1 + r \quad \text{for all } n.$$

Proof: The market is viable if and only if $\tilde{S}_n$ is a martingale, i.e.

$$\begin{aligned} \text{The market is viable} \Leftrightarrow \mathbb{E}^*\left[\tilde{S}_{n+1} | \mathcal{F}_n\right] &= \tilde{S}_n && \text{for all } n \\ \Leftrightarrow \mathbb{E}^*\left[\frac{\tilde{S}_{n+1}}{\tilde{S}_n} \mid \mathcal{F}_n\right] &= 1 && \text{for all } n \\ \Leftrightarrow \mathbb{E}^*\left[\frac{S_{n+1}}{S_n} \mid \mathcal{F}_n\right] &= 1 + r && \text{for all } n. \end{aligned}$$

□

By “conditional probability” $\mathbb{P}(A \mid \mathcal{F}_n)$ of an event A we mean the random variable $\mathbb{E}[\mathbb{1}_A \mid \mathcal{F}_n]$.

We can also evaluate the left-hand side above

$$\begin{aligned}\mathbb{E}^*\left[\frac{S_{n+1}}{S_n} \mid \mathcal{F}_n\right] &= \mathbb{E}^*\left[u\mathbb{1}_{T_n=u} + d\mathbb{1}_{T_n=d} \mid \mathcal{F}_n\right] \\ &= u \cdot \mathbb{E}^*\left[\mathbb{1}_{T_n=u} \mid \mathcal{F}_n\right] + d \cdot \mathbb{E}^*\left[\mathbb{1}_{T_n=d} \mid \mathcal{F}_n\right] \\ &= u \cdot \mathbb{P}^*(T_n = u \mid \mathcal{F}_n) + d \cdot \mathbb{P}^*(T_n = d \mid \mathcal{F}_n)\end{aligned}$$

Putting $p_n = \mathbb{P}^*(T_n = u \mid \mathcal{F}_n)$, this becomes

$$\mathbb{E}^*[T_n|\mathcal{F}_n] = up_n + d(1-p_n).$$

We know the value of this for a viable market, so we can solve for p_n in that case:

Proposition 7.2: The market in this model is viable if and only if

$$\mathbb{P}^*(S_{n+1} = uS_n \mid \mathcal{F}_n) = \frac{1+r-d}{u-d} \text{ for all } n.$$

Proof: By the previous proposition, the market is viable if and only if

$$\begin{aligned} &\mathbb{E}^*[T_n \mid \mathcal{F}_n] = 1+r \\ \text{i.e. iff}\quad & up_n + d(1-p_n) = 1+r \\ \text{i.e. iff}\quad & p_n = \frac{1+r-d}{u-d}\end{aligned}$$

for all n, i.e.

$$\text{The market is viable} \iff \mathbb{P}^*(S_{n+1} = uS_n|\mathcal{F}_n) = \frac{1+r-d}{u-d}.$$

□

Put

$$p := \frac{1+r-d}{u-d}.$$

Remark 7.1: The market is viable if and only if the conditional probability p_n is the particular constant p. That is, the probability does not change whatever knowledge we have from $\mathcal{F}_n$. In particular, all consecutive T_n are independent and $\mathbb{P}(T_n = u|\mathcal{F}_n) = \mathbb{P}(T_n = u)$.

It turns out that the converse is also true:

Proposition 7.3:

$$\forall n : \mathbb{P}(T_n = u \mid \mathcal{F}_n) = \frac{1+r-d}{u-d} \quad \text{if and only if} \quad \begin{array}{c}\forall n : T_n \perp T_{n+1} \\ \text{and } \forall n : \mathbb{P}(T_n = u) = \dfrac{1+r-d}{u-d}\end{array}$$

Proof: We write

$$\mathbb{P}(T_{n+1} = u \mid T_n = u) = \mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid T_n = u\right]$$

and by $\sigma(T_n) \subset \mathcal{F}_{n+1}$ we can condition on $\mathcal{F}_{n+1}$ either before or after the $T_n = u$ condition:

Assuming $\mathbb{P}(T_{n+1} \mid \mathcal{F}_n) = \mathbb{P}(T_{n+1})$ for all n,	Assuming $T_{n+1} \perp T_n$ for all n,
$\begin{aligned} &\mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid T_n = u\right] \\ &= \mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid \mathcal{F}_n \mid T_n = u\right] \\ &= \mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid \mathcal{F}_n\right] \\ &= \mathbb{P}(T_{n+1} = u \mid \mathcal{F}_n) \\ &= \mathbb{P}(T_{n+1} = u) \end{aligned}$	$\begin{aligned} &\mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid T_n = u\right] \\ &= \mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid T_n = u \mid \mathcal{F}_n\right] \\ &= \mathbb{E}\left[\mathbb{1}_{T_{n+1}=u} \mid \mathcal{F}_n\right] \\ &= \mathbb{P}(T_{n+1} = u \mid \mathcal{F}_n), \end{aligned}$
so $\mathbb{P}(T_{n+1} \mid \mathcal{F}_n) = \mathbb{P}(T_{n+1})$ implies $T_{n+1} \perp T_n$.	so $T_{n+1} \perp T_n$ implies $\mathbb{P}(T_{n+1} = u) = \mathbb{P}(T_{n+1} = u \mid \mathcal{F}_n)$

That is,

$$T_{n+1} \perp T_n \quad \text{if and only if} \quad \mathbb{P}(T_{n+1} = u) = \mathbb{P}(T_{n+1} \mid \mathcal{F}_n).$$

The statement follows. □

Joining the equivalences from the last two propositions we get

Proposition 7.4: The following are equivalent:

1. The market is viable;
2. The following conditional probability is the constant p,
$$\mathbb{P}(T_{n+1} | \mathcal{F}_n) = \frac{1 + r - d}{u - d} \text{ for all } n;$$
3. The multiplicative increments T_n are pairwise independent and equal *up* with probability p, i.e.
$$T_{n+1} \perp T_n \quad \text{and} \quad \mathbb{P}(T_n = u) = \frac{1 + r - d}{u - d} \quad \text{for all } n.$$